world in any way, then we're astonishingly lucky that our theories work so well!

Here's an example. Lighting a magnesium sparkler and sticking it into a mixture of powdered rust and aluminum triggers a runaway chemical reaction that quickly reaches about 2,500ºC—nearly half as hot as the surface of the Sun—giving off a blinding light and bringing the iron and aluminum close to their boiling points. This is called the thermite reaction; it is weird and dangerous (seriously, never do this!) but it gets even weirder. Not only is the thermite reaction amazingly intense, but it continues to run until the rust and aluminum are used up, no matter what you do to it. You can put it underwater, you can cover it with sand, you can even put it in the vacuum of space—it will keep burning. (In fact, one of the main industrial applications of thermite is underwater welding.) This is because the reaction doesn't need anything other than the rust and the aluminum, along with a little heat to get it started (hence the sparkler).

The thermite reaction works because aluminum desperately wants to react with oxygen. Rust is nothing but iron and oxygen, so the aluminum rips the oxygen off of the rust, leaving you with aluminum oxide, iron, and a tremendous amount of heat. Or, at least, that's what quantum chemistry has to say on the matter. But if you want to be an instrumentalist, that explanation isn't the real answer. There is no "real" answer to be had. All you care about is the fact that quantum chemistry correctly predicts that you'll get a violent reaction from sticking a magnesium sparkler into a pile of rust and aluminum. The deeper explanation that quantum chemistry gives—why aluminum wants to bond with oxygen so desperately, which has to do with electron orbitals—not only isn't of interest, but isn't real.

But if quantum chemistry's account of thermite isn't real, then the instrumentalist has a serious problem. The theory doesn't just predict that the thermite reaction will happen—it also predicts, in great detail, how it happens and what happens. It tells you how hot the magnesium sparkler has to be to set off the reaction. It tells you exactly how hot the reaction will become, and how long it will last. It even tells you other kinds of rust you can use (different metal oxides) along with the aluminum, and exactly how those will alter the reaction. And all of these excruciatingly detailed answers given by quantum chemistry, down to the fifth decimal place, are explained by the behavior of electron orbitals in the constituent atoms of the